



University of Alkafeel, College of Dentistry, Year 3, Course 1

Pathology Lab 9 Neoplasia

By. Assistant Lecturer, Soroor M. Hadi

Learning Objectives

- ☐ **Define neoplasia.**
- ☐ **Describe the causes of neoplasia.**
- ☐ **Identify the key characteristics of neoplastic cells.**
- ☐ **Classify neoplasms into benign and malignant .**
- ☐ **Explain the mechanisms of invasion and metastasis**

Introduction

Neoplasia is the process of abnormal, uncontrolled proliferation of cells that persists even after the initiating stimulus has been removed.

The resulting mass of tissue is called a **neoplasm (tumor)**.

Causes of Neoplasia

- 1. Genetic factors** (inherited mutations)
- 2. Chemical carcinogens** (e.g., tobacco smoke)
- 3. Physical agents** (radiation)
- 4. Oncogenic viruses** (e.g., HPV)
- 5. Chronic inflammation**

Characteristics of Neoplastic Cells

1.Uncontrolled proliferation

Loss of normal cell cycle regulation.

2.Autonomous growth

Growth continues independent of physiological growth signals.

3.Genetic alterations

Mutations in:

- 1.Proto-oncogenes → oncogenes
- 2.Tumor suppressor genes
- 3.DNA repair genes

4.Clonality

Most tumors arise from a single transformed cell.

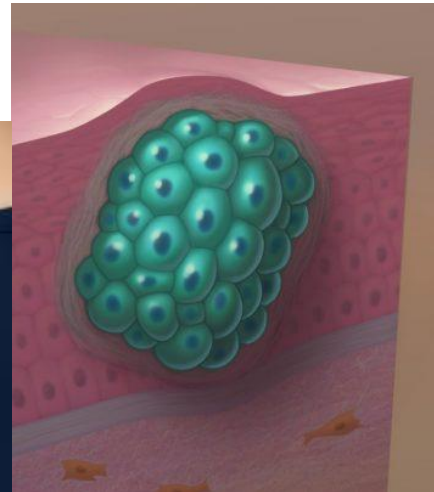
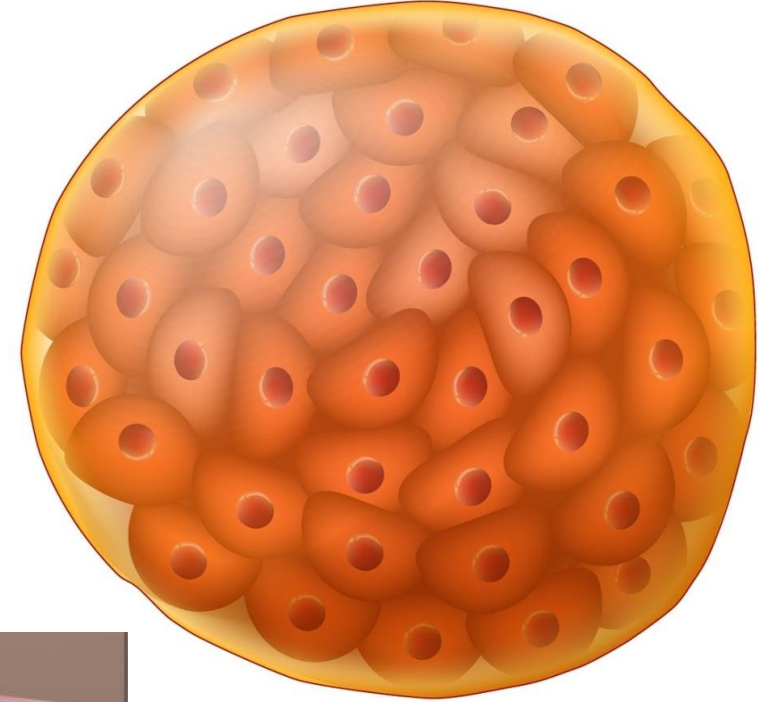
Neoplasms

1. Benign Neoplasms

- Slow growing
- Well-differentiated
- Localized (do not invade or metastasize)
- Usually encapsulated

Examples:

- Lipoma
- Fibroma



BENIGN TUMOR

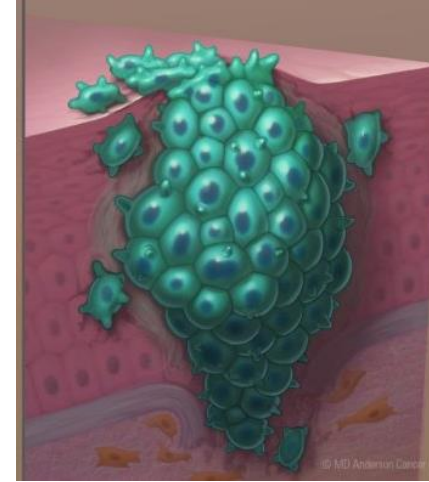
Neoplasms

2. Malignant Neoplasms (Cancer)

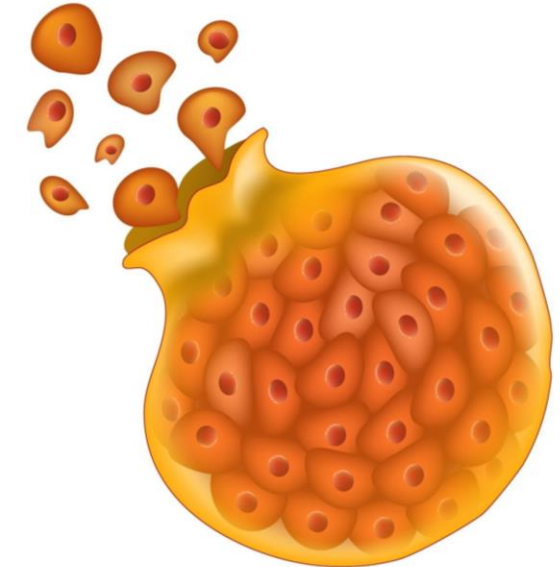
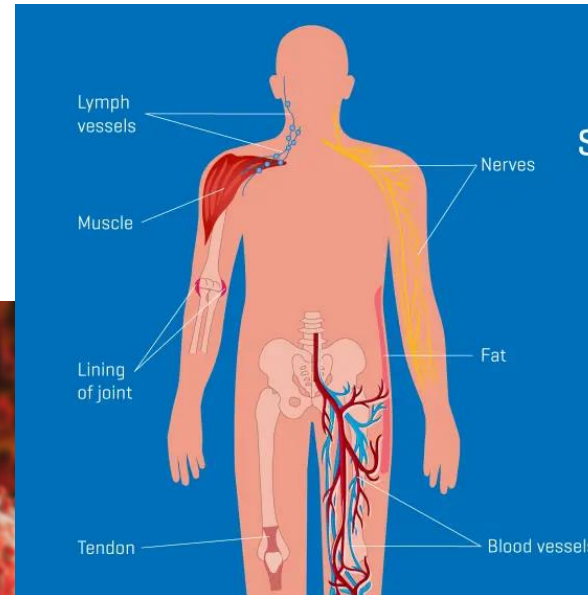
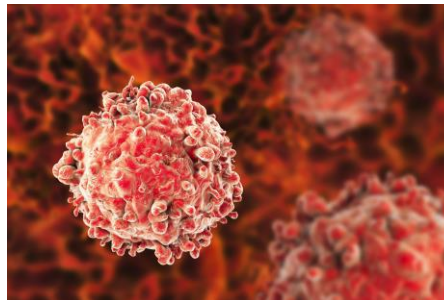
- Rapid growth
- Poor differentiation (anaplasia)
- Invade surrounding tissue
- Can metastasize

Examples:

- Carcinoma
- Sarcoma
- Leukemia



Squamous Cell Carcinoma



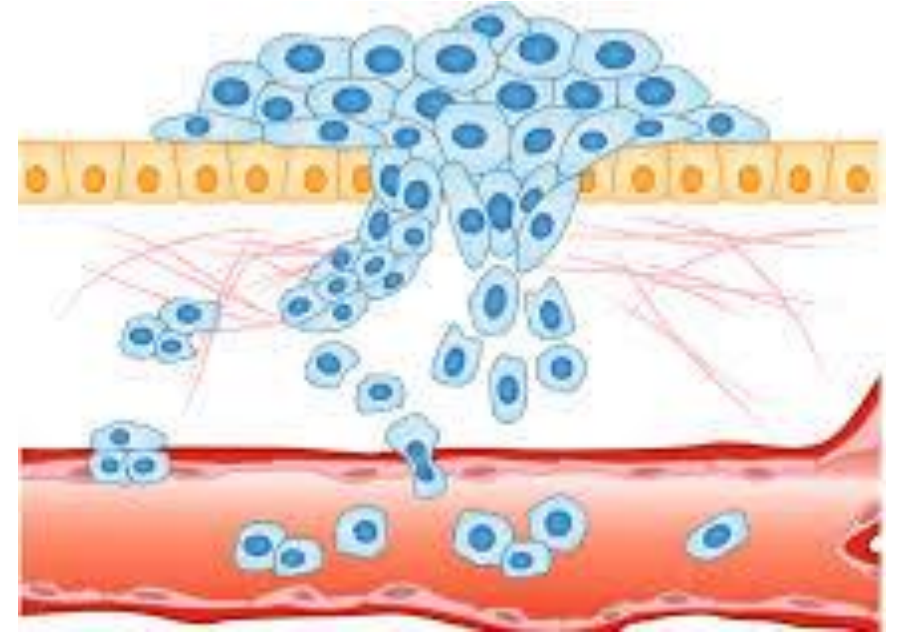
MALIGNANT TUMOR

Invasion

Direct extension and penetration by tumor cells into neighboring tissues.

Ability of cells to break through basement membrane and spread:

- Direct invasion - into surrounding tissue.
- Into lymphatic/vascular channels.
- This is characteristic of malignant cells.



Invasion

1. Loss of Cell-to-Cell Adhesion

Normal epithelial cells adhere via **E-cadherin**.

In carcinoma:

- ↓ E-cadherin expression
- Cells detach from each other

Result → Tumor cells become mobile.

2. Degradation of Extracellular Matrix (ECM)

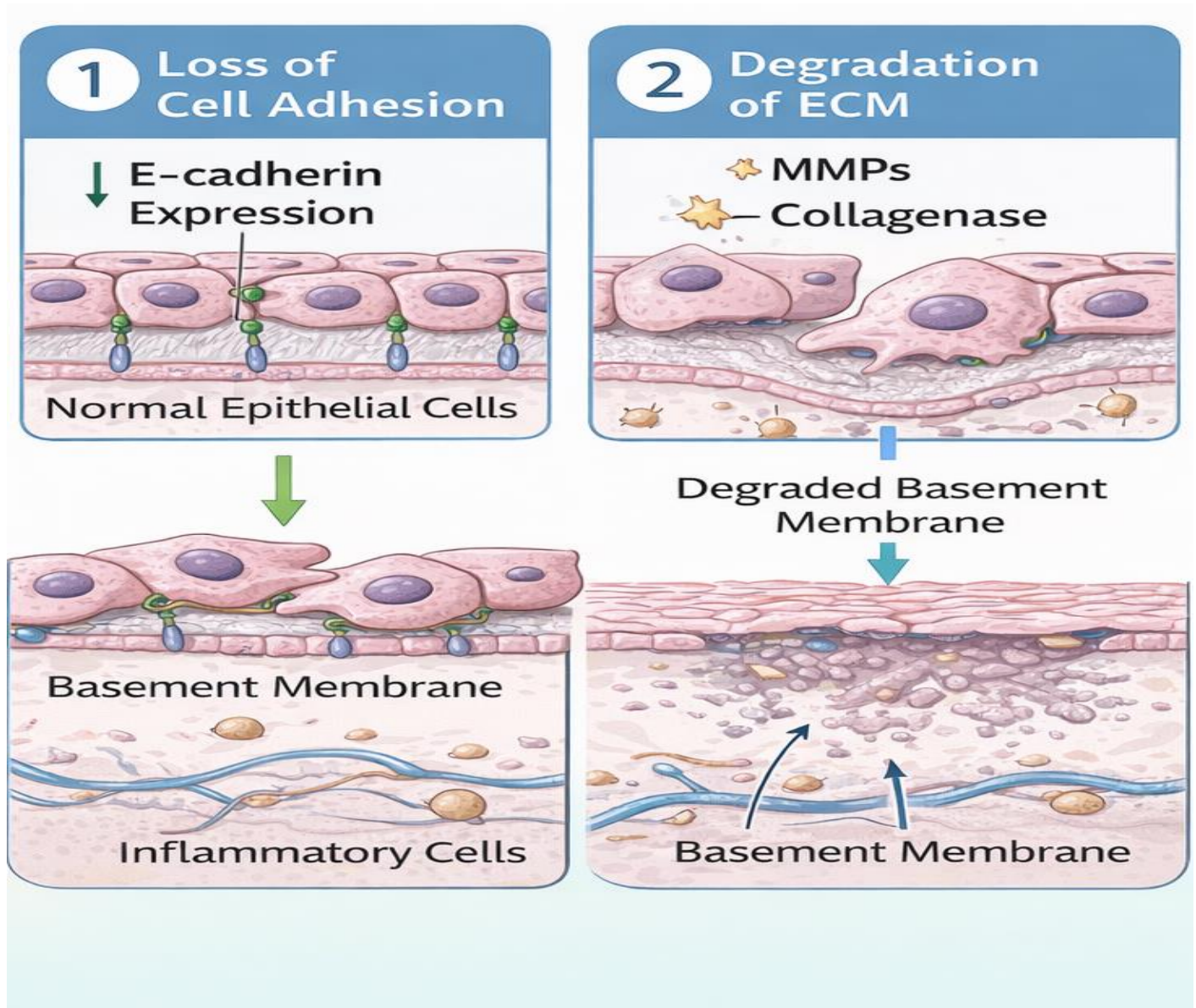
Tumor cells secrete:

- Matrix metalloproteinases (MMPs)
- Collagenases
- Proteases

These enzymes:

- Destroy basement membrane
- Break down collagen

This creates a path for invasion.



Invasion

3. Attachment to ECM Components

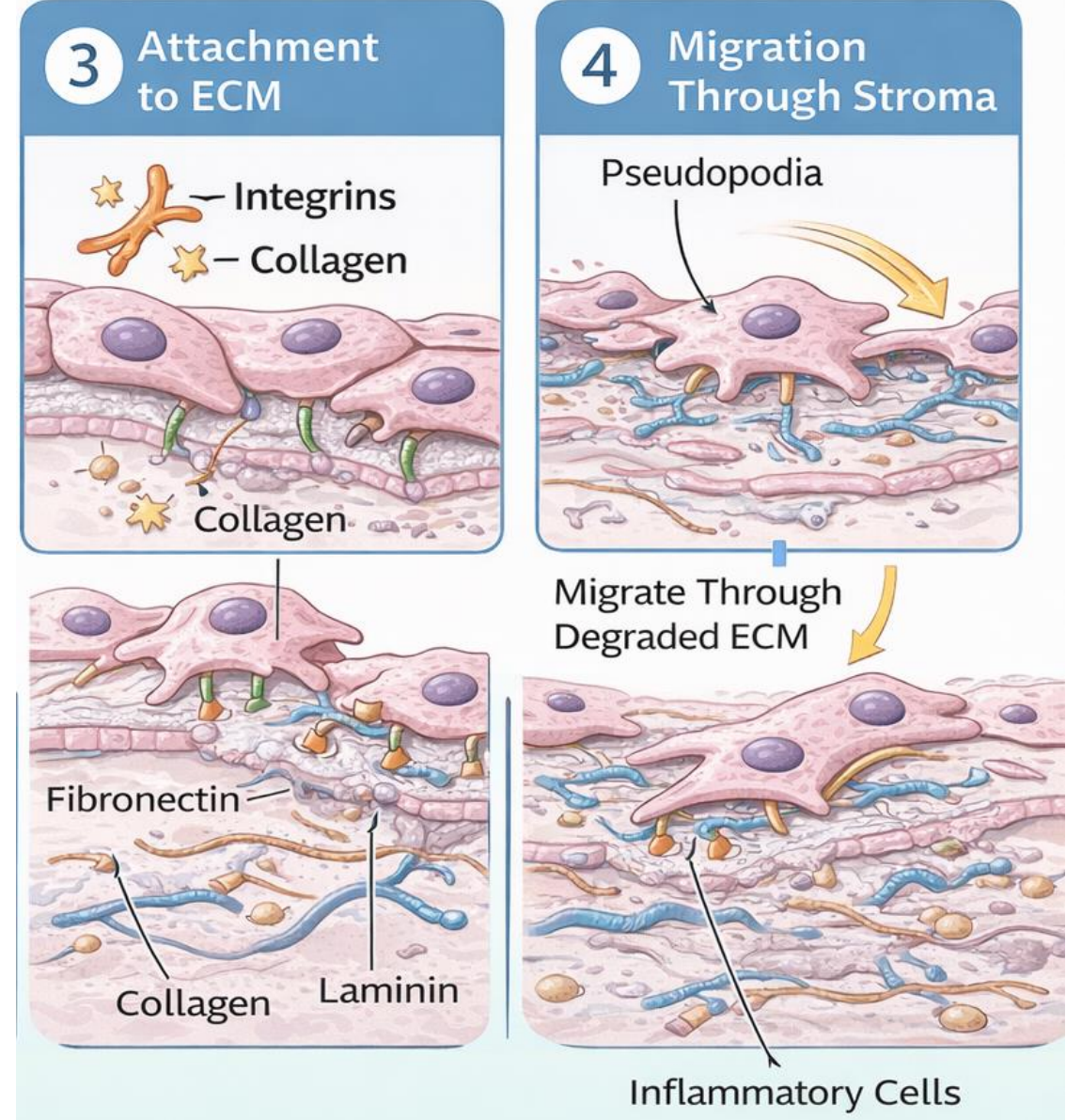
Tumor cells bind to:

- Laminin
- Fibronectin
- Collagen

Through integrin receptors.

4. Migration Through Stroma

Cells actively move through degraded matrix toward blood vessels or lymphatics



Metastasis

Metastasis is the spread of malignant tumor cells from the primary site to a distant site where they form secondary tumors.

Benign tumors DO NOT metastasize.

Routes of Metastasis

1-Lymphatic Spread

Common in carcinomas.

Example:

Breast carcinoma → axillary lymph nodes.

2-Hematogenous Spread

Common in sarcomas.

Tumor cells enter veins → travel to:

- Liver (portal circulation)
- Lungs (systemic circulation)

Example:

Colon carcinoma → liver metastasis.

3-Transcoelomic Spread

Across body cavities.

Example:

Ovarian carcinoma → peritoneal implants.

